Parallel Operation of Integrated Battery Chargers for All Wheel Drive Electric Vehicles

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***Abstract*—In this article, an integrated charging scheme for All Wheel Drive (AWD) Electric Vehicles (EVs) is proposed. The proposed scheme utilizes the existing driving hardware for charging the battery pack with minimum external components. As the driving hardware of AWD EVs includes two motors and two inverters, an Integrated Battery Charger (IBC) with parallel configuration is proposed. In the existing topology of three-phase IBC using a split-phase machine, the grid currents were not balanced. But in the proposed parallel IBC topology, the two split phase machines are proposed with different phase configurations, namely R-Y-B and Y-R-B, which will cancel the unbalance caused by each IBC, and make the grid currents balanced. The mathematical proof for the same is also included in this article. The proposed parallel topology of IBCs has a higher power rating and hence provides faster charging compared to conventional single IBCs. Circulating current mitigation for parallel IBCs is also presented in this article. Exhaustive simulation and experimental results are included to validate the proposed scheme.**

***Index Terms*—Electric vehicle (EV) charging, fast on-board charger, integrated battery charger (IBC), parallel inverters, split-phase machine.**

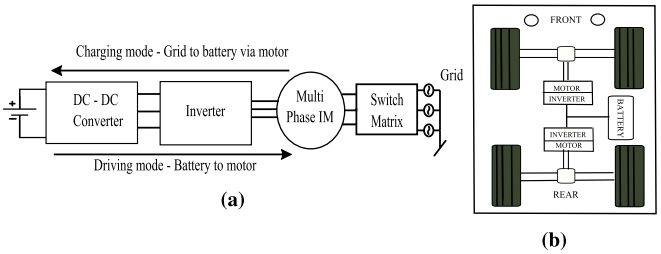

Fig. 1. (a) IBC [3]. (b) Power train of an AWD.

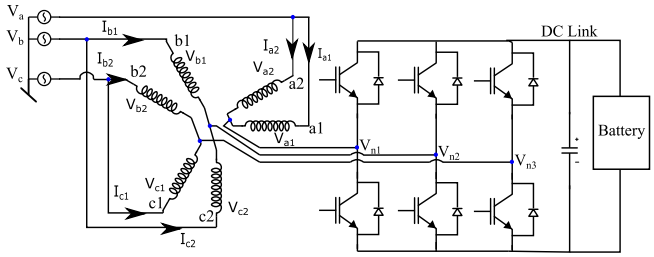

Fig. 2. Schematic of IBC using a split-phase machine [8].

I. Introduction

THE battery charging technology is one of the key areas of research in the electric vehicle (EV) industry. One of the main goals of all automotive companies around the world is to increase the range of the vehicle over a single charge. Also, it has become the need of the hour to enable charging the battery as quickly as possible that too using its on-board charger.

Among various types of onboard chargers, integrated battery charger (IBC) has the advantage of providing faster charging using the existing high power driving hardware within the vehicle [1], [2]. As the name suggests, an IBC utilizes the same high power driving hardware of the vehicle for charging the battery [see Fig. 1(a)] making it a fast charger [3].

The only constraint for an IBC is that the motor should not rotate while charging. The first IBCs that were developed, used single phase supply [4]. However, these single phase IBCs had a very slow charging rate. Thus IBCs using three phase supply were developed using different types of motors like induction motors, Permanent Magnet Synchronous machines (PMSM) [5], switched reluctance motors (SRM) [6] etc. But the three phase IBCs developed using three phase motors required additional hardware to block the rotor while charging the battery. With the development of multiphase machines, IBCs were developed using these machines [7]. But this required the replacement of existing three phase inverters with multiphase inverters.

In [8], an IBC was developed with split phase machine. The advantage of using IBC with split phase machine is that the three phase machine and three legged inverter that is already present in the power-train of the EV can be reutilized instead of replacing them with a multiphase machine and multi legged inverter. The stator phase belt of three phase machine can be divided equally to form two sets of three phase windings shifted spatially by 30° resulting into the winding configuration of split phase machine [9]. The winding configuration of the motor during charging mode and driving mode are discussed in detail in [8].

The schematic of the IBC using split phase machine is shown in Fig. 2. The inverter acts as a front-end converter and ensures that the power is drawn from the grid at unity power factor. The windings of the machine are connected to the grid in such a way that only a pulsating flux is developed within the stator, hence the instantaneous torque is zero. However, the stator windings of the split phase machine is unsymmetrical, hence the emf induced in each winding due to the presence of the pulsating magnetic flux is different from each other. This

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asymmetry creates an unbalance in the three-phase currents drawn from the grid. The mathematical model of this system is developed in [10] and new control methods were implemented to limit the unbalance to a large extent.

In this article, a parallel combination of two IBCs with split-phase machine is proposed. This can be implemented in All Wheel Drive (AWD) EVs. AWD vehicle is capable of providing power to all four wheels continuously. AWD EVs consist of two sets of inverter and motor as shown in Fig. 1(b). They have independent control of front wheel and back wheels, prevents wheel lock and slip, and hence have good driving ability on off-roads, snowy roads, and wet roads.

As an IBC, the two motor drives of the AWD can be connected in parallel to form a fast charger since the power will be twice that of a single IBC using split phase machine.

In this article, a new configuration for connecting two IBCs using split phase machine in parallel is proposed. The split phase machine has an unsymmetrical structure. Also the field induced inside the machine is pulsating in nature. Hence the currents were found to be unbalanced when the split phase machine was used for single phase IBC. In the proposed topology, there are two split phase machines. Each of the split phase machine can be connected in six different configurations namely, R-Y-B, R-B-Y, Y-B-R, Y-R-B, B-R-Y and B-Y-R with the grid. It is proposed in this work that when the first split phase machine is connected in R-Y-B configuration and second split phase machine is connected in Y-R-B configuration, the unbalance that occurs in each of the single phase IBCs is canceled when connected in parallel and the grid current is found to be balanced. Mathematical proof for the proposed configuration is also included which clearly shows the cancelation of the unbalance.

Circulating currents that are inherent to parallel converter systems is mitigated by adopting zero-sequence current-based modulation technique. Also by having two IBCs in parallel a much faster charging could be obtained due to higher power rating. The proposed parallel configuration of IBCs is discussed in detail along with its model in Section II. Section III derives the zero sequence model of the system and the issue of circulating current through the parallel converter system. The control structure of the system is given in Section IV. Section V gives the simulation results, and Section VI provides the experimental results.

II. Parallel Operation of IBC Using a Split Phase Machine

The proposed parallel combination of two IBCs is shown in Fig. 3. The two IBCs, namely IBC1 and IBC2 are connected in parallel between the three phase supply and the battery. In this article, the windings of the split phase motor in IBC1 and IBC2 are configured in different ways to completely eliminate the unbalance that was present in a single IBC. The windings of the motor in IBC1 is given R-Y-B configuration and windings of the motor in IBC2 is given Y-R-B configuration, which makes the parallel system a balanced system. The following subsections give a detailed description of the system.

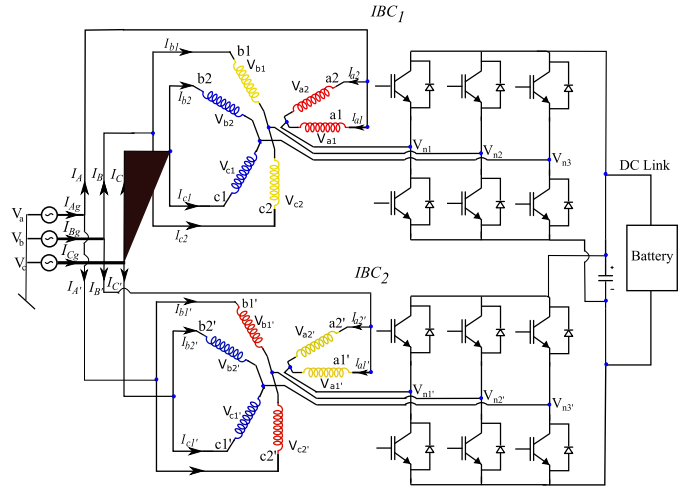

Fig. 3. Proposed configuration of parallel IBCs (R-Y-B and Y-R-B phase sequences for IBC1 and IBC2, respectively).

A. Model and Analysis of IBC1

The split phase machine's winding configuration of IBC1 is shown in Fig. 4(a). The windings of the split phase machine can be considered as two sets of three phase windings which are phase shifted by 30°. The first set of windings a_1–c_1 are given voltages V_a–V_c, respectively, which produces a rotating flux in one direction and the second set of windings a_2–c_2 are given voltages V_a, V_c, V_b, respectively, creating a rotating flux in the opposite direction. Hence the net flux will be pulsating in nature as shown in Fig. 4(a) and the machine will not rotate while charging. In this article the sequence of grid supply given to the split phase machine of IBC1 is R-Y-B. As seen from Fig. 4(a), the parallel combination of a_1 and a_2 windings act as an interface inductance between the grid and the inverter for "a" phase, similarly the parallel combination of b_1 and c_2 windings will be the interface inductance of "b" phase, and the parallel combination of c_1 and b_2 will be the interface inductance of "c" phase. Due to the unsymmetrical nature of the interface inductances connected to three phases of the grid, the grid currents tend to be unbalanced. The grid currents are given by the following equation:

$$I_A = I_{a_1} + I_{a_2}; \quad I_B = I_{b_1} + I_{c_2}; \; I_C = I_{c_1} + I_{b_2} \tag{1}$$

where, I_A–I_C are the grid currents of IBC1 and $I_{a_1}, I_{a_2}, I_{b_1}, I_{b_2}, I_{c_1}$ and I_{c_2} are the currents flowing through the individual split phase motor windings a_1, a_2, b_1, b_2, c_1 and c_2, respectively [see Fig. 4(a)]. The mathematical model of IBC using split phase machine is derived in [10] where the unbalance is theoretically expressed in the inductance matrix of the system. The same model is used in the present article for obtaining the inductance matrix of IBC1 and IBC2.

The stationary and synchronous rotating frames for modeling IBC1 is shown in Fig. 4(b). $\alpha-\beta$ is the stationary reference frame, and $d_m - q_m$ is the synchronously rotating frame that rotates with the grid frequency ω. The α-axis is aligned along the direction of pulsating magnetic field.

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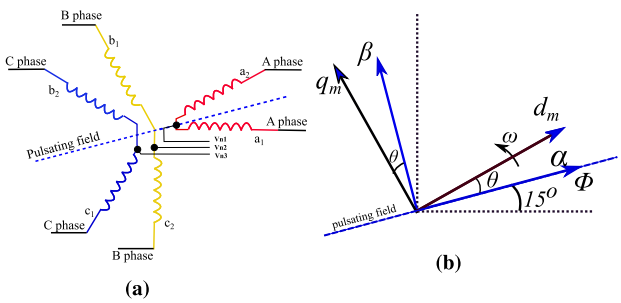

Fig. 4. IBC1: R-Y-B configuration and reference frame. (a) R-Y-B winding configuration of split phase machine. (b) Reference frame definition: stationary and synchronous reference frame.

The motor phase voltages are transformed into the defined reference frames using following equations:

$$\begin{bmatrix} V_{\alpha 1} \\ V_{\beta 1} \end{bmatrix} = \begin{bmatrix} \cos(15^\circ) & \cos(105^\circ) & \cos(225^\circ) \\ \cos(105^\circ) & \cos(15^\circ) & \cos(135^\circ) \end{bmatrix} \begin{bmatrix} V_{a_1} \\ V_{b_1} \\ V_{c_1} \end{bmatrix} \quad (2)$$

$$\begin{bmatrix} V_{\alpha 2} \\ V_{\beta 2} \end{bmatrix} = \begin{bmatrix} \cos(15^\circ) & \cos(135^\circ) & \cos(255^\circ) \\ \cos(75^\circ) & \cos(45^\circ) & \cos(165^\circ) \end{bmatrix} \begin{bmatrix} V_{a_2} \\ V_{b_2} \\ V_{c_2} \end{bmatrix}. \quad (3)$$

Here, $V_{\alpha 1}, V_{\beta 1}$ are voltages corresponding to windings a_1–c_1 represented in the stationary reference frame and $V_{\alpha 2}, V_{\beta 2}$ are voltages corresponding to windings a_2–c_2. Since the windings are connected in parallel between the grid and inverter legs, the voltage across the phase windings are given by the following equation:

$$V_{a_1} = V_{a_2} = V_a; \quad V_{b_1} = V_{c_2} = V_b; \quad V_{c_1} = V_{b_2} = V_c. \quad (4)$$

Similarly, the currents through the windings a_1–c_1 and a_2–c_2 can be transformed into the stationary reference frame as $I_{\alpha 1}, I_{\beta 1}$ and $I_{\alpha 2}, I_{\beta 2}$ respectively. It can be proved that

$$I_{\alpha 1} = I_{\alpha 2}; \quad I_{\beta 1} = -I_{\beta 2}. \quad (5)$$

Also, by further analysis [10], it is found that during charging mode of operation, the current from grid in each phase is equally divided between the corresponding parallel windings

$$I_{a1} = I_{a2}; \quad I_{b1} = I_{c2} \text{ and } I_{c1} = I_{b2}. \quad (6)$$

The conditions derived by (5) and (6) verifies that the machine flux is linked only along the α-axis. Hence the stator and rotor β-axis component of current will be zero [10]. Hence the instantaneous electromagnetic torque generated inside the machine for the IBC will be zero as follows:

$$\tau = \frac{2}{3} L_{sr} \left(I_\beta I_{r\alpha} - I_\alpha I_{r\beta} \right) = 0. \quad (7)$$

Fig. 5 shows the experimental result verifying that the torque generated by the split-phase machine is zero when configured as an IBC. The equations to transform stationary reference frame into synchronous reference frame [11] is given in the following equations:

$$I_{d_{g1}} + j I_{q_{g1}} = \left(I_{\alpha 1} + j I_{\beta 1} \right) e^{-j\omega t} \quad (8)$$

$$I_{d_{g2}} + j I_{q_{g2}} = \left(I_{\alpha 2} + j I_{\beta 2} \right) e^{j\omega t} \quad (9)$$

$$I_{rd_g} + j I_{rq_g} = \left(I_{r\alpha} + j I_{r\beta} \right) e^{-j\omega t} \quad (10)$$

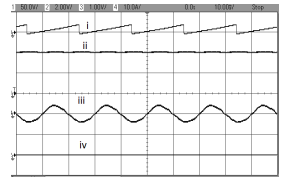

Fig. 5. Experimental result. i: grid angle θ, ii: dc-link voltage (50 V/div), iii: phase current (10 A/div), and iv: torque estimated from DAC [8].

where, $I_{d_{g1}}, I_{q_{g1}}$ and $I_{d_{g2}}, I_{q_{g2}}$ are the stator currents in rotating reference frame corresponding to the currents flowing through a_1–c_1 and a_2–c_2 windings, respectively. Similarly I_{rd_g}, I_{rq_g} represent rotor current vectors in the rotating reference frame. The input phase currents to IBC1, namely I_A–I_C are transformed into I_{d_m} and I_{q_m} as follows:

$$I_{d_m} = I_{d_{g1}} + I_{d_{g2}}; \quad I_{q_m} = I_{q_{g1}} - I_{q_{g2}}. \quad (11)$$

1) Inductance Model and Unbalance in IBC1: The mathematical model of IBC using split phase machine in rotating reference frame was derived in [10]. An inductance matrix was derived for stator voltages considering only the stator currents as is given in the following equation:

$$\begin{bmatrix} V_{d_m} \\ V_{q_m} \end{bmatrix} = \begin{bmatrix} L_d \dfrac{d}{dt} - \omega M_1 & -\omega L'_{ss} - \omega M_2 - L_{dq} \dfrac{d}{dt} \\ \omega L'_{ss} - \omega M_2 - L_{dq} \dfrac{d}{dt} & L_q \dfrac{d}{dt} + \omega M_1 \end{bmatrix} \times \begin{bmatrix} I_{d_m} \\ I_{q_m} \end{bmatrix} \quad (12)$$

where, L_{ss} (stator inductance per phase); $L'_{ss} = (L_{ss}/2)$; L_d (inductance in d-axis) $= L'_{ss} + L'_{ss}(1 - \sigma_{12}) \cos(2\omega t)$; L_q (inductance in q-axis) $= L'_{ss} - L'_{ss}(1 - \sigma_{12}) \cos(2\omega t)$; $M_1 = L'_{ss}(1 - \sigma_{12}) \sin(2\omega t)$; $M_2 = L'_{ss}(1 - \sigma_{12}) \cos(2\omega t)$ L_{dq} (Inductance between dq-axis) $= L'_{ss}(1 - \sigma_{12}) \sin(2\omega t)$; $(1 - \sigma_{12}) L_{ss}$ = stator mutual inductance (coupling coefficient) between split phase stator groups and ω is the frequency of rotation of the synchronous reference frame.

From the inductance matrix of the IBC, the inductance terms are found to be varying with the grid frequency. There exists a mutual inductance term L_{dq}, between the orthogonal rotating d- and q-axis which will be absent for a balanced system. Also the d-axis inductance term and q-axis inductance terms are different, wherein for a balanced system they are equal. Hence from the above matrix, it can be seen that the presence of pulsating field on the unsymmetrical windings of split phase machine makes the system unbalanced.

B. Modeling and Analysis of IBC2

Fig. 6(a) shows the winding configuration of split phase windings of IBC2 when in charging mode. The grid voltage applied to windings a'_1–c'_1 is V_b, V_a, V_c, respectively, and the voltage applied to windings a'_2–c'_2 is V_b, V_c, V_a, respectively, resulting in a net pulsating flux inside the motor. This will make sure that the instantaneous torque inside the machine is zero when in charging mode. Modeling of IBC2 in stationary reference frame and synchronously rotating frame is done similar to the procedure provided for IBC1 [12]. The following

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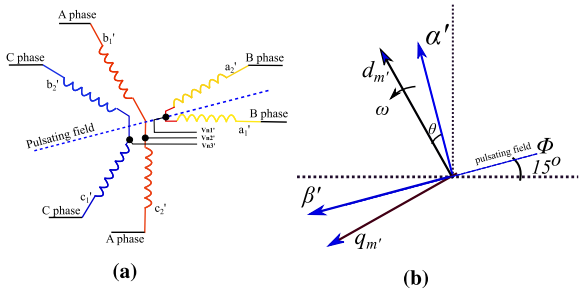

Fig. 6. IBC2: Y-R-B winding configuration and reference frame. (a) Y-R-B winding configuration of split phase machine. (b) dq reference frame for IBC2.

equations gives the transformation of phase voltages of IBC2 to the corresponding voltages in stationary reference frames:

$$\begin{bmatrix} V_{\alpha 1'} \\ V_{\beta 1'} \end{bmatrix} = \begin{bmatrix} \cos(15^\circ) & \cos(105^\circ) & \cos(225^\circ) \\ \cos(105^\circ) & \cos(15^\circ) & \cos(135^\circ) \end{bmatrix} \begin{bmatrix} V_{a_1'} \\ V_{b_1'} \\ V_{c_1'} \end{bmatrix} \quad (13)$$

$$\begin{bmatrix} V_{\alpha 2'} \\ V_{\beta 2'} \end{bmatrix} = \begin{bmatrix} \cos(15^\circ) & \cos(135^\circ) & \cos(255^\circ) \\ \cos(75^\circ) & \cos(45^\circ) & \cos(165^\circ) \end{bmatrix} \begin{bmatrix} V_{a_2'} \\ V_{b_2'} \\ V_{c_2'} \end{bmatrix}. \quad (14)$$

Here, $V_{\alpha 1'}$, $V_{\beta 1'}$ and $V_{\alpha 2'}$, $V_{\beta 2'}$ are the voltages transformed to stationary reference frames corresponding to voltages across windings a_1'–c_1' and a_2'–c_2', respectively. As seen in Fig. 6(a) the voltage across the windings for Y-R-B phase sequence is given by the following equation:

$$V_{a_1'} = V_{a_2'} = V_{b'}; \quad V_{b_1'} = V_{c_2'} = V_{a'}; \quad V_{c_1'} = V_{b_2'} = V_{c'}. \quad (15)$$

Similarly, the currents through the windings a_1'–c_1' and a_2'–c_2' can be transformed into the stationary reference frame as $I_{\alpha 1'}, I_{\beta 1'}$ and $I_{\alpha 2'}, I_{\beta 2'}$ respectively. When the mathematical model of IBC1 and IBC2 are compared using (2)–(4), (13)–(15), it can be seen that the variables in α'-axis of IBC2 corresponds to the voltages in β-axis of IBC1. Similarly, the variables in β'-axis of IBC2 corresponds to variables in $-\alpha$-axis of IBC1. In other words, the stationary reference frame $\alpha' - \beta'$ of IBC2 is shifted by 90° from stationary frame $\alpha - \beta$ of IBC1. The reference frames of IBC2 is shown in Fig. 6(b). The equations to transform from stationary reference frame to synchronous reference frame for IBC2 is given by (16)

$$I_{d_{g1'}} + jI_{q_{g1'}} = \left(I_{\alpha 1'} + jI_{\beta 1'}\right)e^{j\omega t} \quad (16)$$

$$I_{d_{g2'}} + jI_{q_{g2'}} = \left(I_{\alpha 2'} + jI_{\beta 2'}\right)e^{-j\omega t} \quad (17)$$

$$I_{rd_{g'}} + jI_{rq_{g'}} = \left(I_{r\alpha'} + jI_{r\beta'}\right)e^{-j\omega t} \quad (18)$$

where $I_{d_{g1}'}, I_{q_{g1}'}$ and $I_{d_{g2}'}, I_{q_{g2}'}$ are the stator currents in rotating reference frames corresponding to the currents through a_1'–c_1' and a_2'–c_2' windings, respectively. Similarly, I_{rd_g}', I_{rq_g}' represent rotor current vectors in rotating reference frame. The input phase currents to IBC2, namely I_A'–I_C' are transformed into I_{d_m}' and I_{q_m}' as shown in the following equation:

$$I_{d_m}' = I_{d_{g1}'} + I_{d_{g2}'}; \quad I_{q_m}' = I_{q_{g2}'} - I_{q_{g1}'}. \quad (19)$$

1) Inductance and Unbalance in IBC2: Inductance matrix of IBC2 is derived in the following equation similar to IBC1:

$$\begin{bmatrix} -V_{q_m'} \\ V_{d_m'} \end{bmatrix} = \begin{bmatrix} L_d \dfrac{d}{dt} - \omega M_1 & -\omega L_{ss}' - \omega M_2 - L_{dq}\dfrac{d}{dt} \\ \omega L_{ss}' - \omega M_2 - L_{dq}\dfrac{d}{dt} & L_q \dfrac{d}{dt} + \omega M_1 \end{bmatrix} \times \begin{bmatrix} -I_{q_m'} \\ I_{d_m'} \end{bmatrix}. \quad (20)$$

Similar to IBC1, IBC2 is also an unbalanced system and the motor input currents, I_A'–I_C' are not purely sinusoidal. Inductance terms are time varying functions related to the grid angle and frequency. Mutual inductance exists between the two orthogonal rotating d- and q-axes.

C. Proposed IBC System for AWD EVs-Parallel Configuration of IBC1 and IBC2

In this article, a parallel combination of IBC1 and IBC2 between the grid and the battery bank is proposed, which can be configured using the existing hardware in an AWD vehicle. To reduce the unbalance in grid currents due to asymmetry in inductances, IBC1 is given R-Y-B phase-sequence and IBC2 is given Y-R-B phase-sequence as shown in Fig. 3. The mathematical model of the parallel system is derived to show the reduction in current unbalance when IBC1 and IBC2 are connected in RYB-YRB parallel configuration.

The grid currents I_{A_g}–I_{C_g} is the sum of the phase currents to IBC1 and IBC2 as given in the following equation:

$$I_{A_g} = I_A + I_{A'}; \quad I_{B_g} = I_B + I_{B'}; \quad I_{C_g} = I_C + I_{C'}. \quad (21)$$

The grid currents in synchronous reference frame, I_{d_g} and I_{q_g} is the sum of the currents in IBC1 and IBC2 as shown in the following equation:

$$I_{d_g} = I_{d_m} + I_{d_m'}; \quad I_{q_g} = I_{q_m} + I_{q_m'}. \quad (22)$$

1) Inductance Matrix of the Parallel System: The inductance matrix of two IBCs are connected in parallel is given by the following equation:

$$\begin{bmatrix} V_{d_m} \\ V_{q_m} \end{bmatrix} + \begin{bmatrix} V_{d_m'} \\ V_{q_m'} \end{bmatrix} = \begin{bmatrix} L_d \dfrac{d}{dt} - \omega M_1 & -\omega L_{ss}' - \omega M_2 - L_{dq}\dfrac{d}{dt} \\ \omega L_{ss}' - \omega M_2 - L_{dq}\dfrac{d}{dt} & L_q \dfrac{d}{dt} + \omega M_1 \end{bmatrix} \begin{bmatrix} I_{d_m} \\ I_{q_m} \end{bmatrix} + \begin{bmatrix} L_q \dfrac{d}{dt} + \omega M_1 & -\omega L_{ss}' + \omega M_2 + L_{dq}\dfrac{d}{dt} \\ \omega L_{ss}' + \omega M_2 + L_{dq}\dfrac{d}{dt} & L_d \dfrac{d}{dt} - \omega M_1 \end{bmatrix} \times \begin{bmatrix} I_{d_m'} \\ I_{q_m'} \end{bmatrix}. \quad (23)$$

It can be seen from (23) that the variable inductance terms that depend upon the angular frequency ω and the grid angle θ of IBC1 is opposite to the corresponding terms in IBC2. Thus, the unbalance in IBC1 is canceled by unbalance in IBC2. Assuming an ideal case when the input currents of IBC1 and

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IBC2 are equal, ie, $I_{d_m} = I_{d'_m}$ and $I_{q_m} = I_{q'_m}$, then (23) reduces to the following equation:

$$\begin{bmatrix} V_{d_m} \\ V_{q_m} \end{bmatrix} + \begin{bmatrix} V_{d'_m} \\ V_{q'_m} \end{bmatrix} = \begin{bmatrix} L'_{\mathrm{ss}} & -\omega L'_{\mathrm{ss}} \\ \omega L'_{\mathrm{ss}} & L'_{\mathrm{ss}} \end{bmatrix} \begin{bmatrix} I_{d_g} \\ I_{q_g} \end{bmatrix}. \tag{24}$$

This represents the inductance matrix of a balanced system. The inductance in d- and q-axis are equal, there is no mutual coupling between the d- and q-axis terms and they are not a function of the grid angle θ. The current passing through two motors will not be same in practice, but maximum cancelation in the unbalance can be achieved by parallel combination of IBC1 with R-Y-B configuration and IBC2 with Y-R-B configuration.

Similarly two more configurations for parallel IBCs are possible which can reduce the unbalance. These are as follows.

1) IBC1: R-B-Y, IBC2: B-R-Y.
2) IBC1: Y-B-R, IBC2: B-Y-R.

III. Circulating Current in the Proposed System

When two converters are connected in parallel with a common dc link, a path exists for a circulating current to flow between the two converters. The circulating current flowing through the system can have positive sequence, negative sequence and zero sequence terms [13]. In this article, zero sequence model of the system is derived to find the behavior of the zero sequence circulating current.

A. Zero Sequence Current of IBC1

The instantaneous voltage across each winding of the split phase machine can be written as in (25), shown at the bottom of the page, where, $L'_{\mathrm{ls}} = (L_{\mathrm{ls}}/2)$ = Leakage inductance of each winding; $L'_{\mathrm{ms}} = (L_{\mathrm{ms}}/2)$ = mutual inductance between two windings and $R' = (R/2)$. Rearranging (25), (26), shown at the bottom of the page, can be derived, where $Z = L'_{\mathrm{ls}}(L'_{\mathrm{ls}} + 3L'_{\mathrm{ms}})$.

The voltage across the winding $a1$, i.e., V_{a1} can be written in terms of the grid voltage and inverter pole voltage as given in (27) where V_{an} is the phase to neutral voltage of the grid, V_{a1o} is the voltage of "a" phase leg of inverter with respect to the common point "O," V_{o1}, V_{o2} is the common mode voltage of IBC1 and IBC2, V_{no} is the voltage of the common point "O" with respect to the neutral

$$V_{a1} = (V_{a1o} - V_{no}) - V_{an} = \left(V_{a1o} - \frac{1}{2}(V_{o1} + V_{o2})\right) - V_{an}. \tag{27}$$

In a similar way, the voltages V_{b1} and V_{c1} can be defined in terms of the common mode voltage. The definition of zero sequence current for IBC1 is given by the following equation:

$$I_{o1} = \frac{I_A + I_B + I_C}{3}. \tag{28}$$

Using the above equations, the common mode current of IBC1 is derived in the following equation:

$$pI_{o1} = \frac{\frac{1}{2}(V_{o1} - V_{o2}) - \frac{R}{2}I_{o1}}{\frac{L_{\mathrm{ls}}}{2}}. \tag{29}$$

B. Zero Sequence Current of IBC2

For IBC2, the zero sequence current I_{o2} can be derived in a similar way, and is given by the following equation:

$$pI_{o2} = \frac{\frac{1}{2}(V_{o2} - V_{o1}) - \frac{R}{2}i_{o2}}{\frac{L_{\mathrm{ls}}}{2}}. \tag{30}$$

From (29) and (30) it can be seen that the common mode current depends mainly on the difference in common mode voltage and leakage inductance of the machine windings and is independent of the varying self and mutual inductances. Also the circulating current equation in (29) is equal and opposite to the circulating current equation in (30). Hence only one zero sequence controller is required to limit the circulating current in the parallel system.

In this parallel system, IBC1 has R-Y-B configuration and IBC2 has Y-R-B configuration. Hence same carriers are used for both the IBC as interleaving will not produce any harmonic suppression. Since the carriers are not interleaved, there wont be any high frequency circulating current present in the system. But there exists a low frequency circulating

$$\begin{bmatrix} V_{a1} \\ V_{b1} \\ V_{c1} \end{bmatrix} = \begin{bmatrix} R' & 0 & 0 \\ 0 & R' & 0 \\ 0 & 0 & R' \end{bmatrix} \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} + \begin{bmatrix} L'_{\mathrm{ls}} + L'_{\mathrm{ms}} + L'_{\mathrm{ms}}\cos(30^\circ) & L'_{\mathrm{ms}}\cos(120^\circ) & L'_{\mathrm{ms}}\cos(150^\circ) + L'_{\mathrm{ms}}\cos(240^\circ) \\ L'_{\mathrm{ms}}\cos(120^\circ) & L'_{\mathrm{ls}} + L_{\mathrm{ms}} - L'_{\mathrm{ms}}\cos(30^\circ) & L'_{\mathrm{ms}}\cos(240^\circ) + L'_{\mathrm{ms}}\cos(330^\circ) \\ L'_{\mathrm{ms}}\cos(120^\circ) + L'_{\mathrm{ms}}\cos(150^\circ) & L'_{\mathrm{ms}}\cos(120^\circ) + L'_{\mathrm{ms}}\cos(30^\circ) & L'_{\mathrm{ls}} + L'_{\mathrm{ms}} \end{bmatrix} \begin{bmatrix} \frac{d}{dt}I_A \\ \frac{d}{dt}I_B \\ \frac{d}{dt}I_C \end{bmatrix} \tag{25}$$

$$\begin{bmatrix} \frac{d}{dt}I_A \\ \frac{d}{dt}I_B \\ \frac{d}{dt}I_C \end{bmatrix} = \frac{1}{Z} \begin{bmatrix} L'_{\mathrm{ls}} + 1.133L'_{\mathrm{ms}} & 0.5L'_{\mathrm{ms}} & 1.366L'_{\mathrm{ms}} \\ 0.5L'_{\mathrm{ms}} & L'_{\mathrm{ls}} + 2.866L'_{\mathrm{ms}} & -0.366L'_{\mathrm{ms}} \\ 1.366L'_{\mathrm{ms}} & -0.366L'_{\mathrm{ms}} & L'_{\mathrm{ls}} + 2L'_{\mathrm{ms}} \end{bmatrix} \begin{bmatrix} V_{a1} - R'I_A \\ V_{b1} - R'I_B \\ V_{c1} - R'I_C \end{bmatrix} \tag{26}$$

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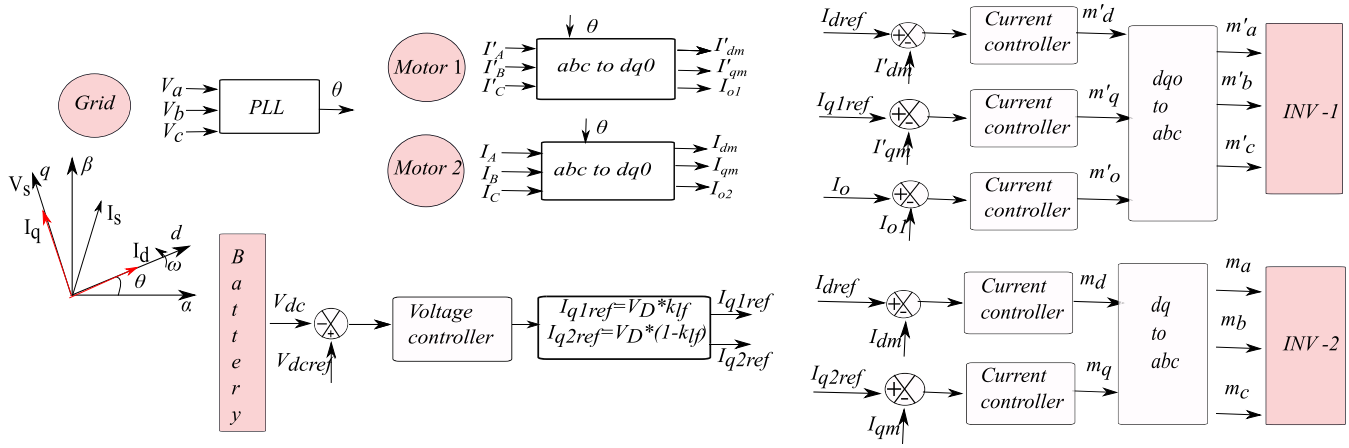
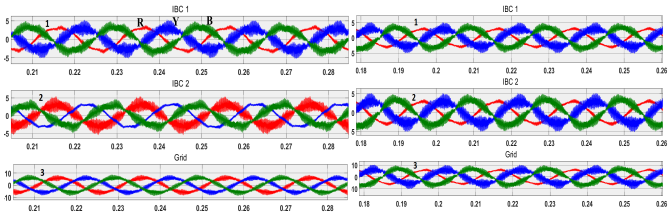


Fig. 7. Control block diagram for parallel IBC.



(a) (b)

Fig. 8. Simulation results: steady state current waveforms: (a) proposed R-Y-B/Y-R-B configuration and (b) R-Y-B/R-Y-B configuration. 1: Three phase input currents (I_A–I_C) for IBC1, 2: three phase input currents ($I_{A'}$–$I_{C'}$) for IBC2, and 3: three phase grid currents (I_{A_g}–I_{C_g}).

current due to the difference in pole voltage of inverters of IBC1 and IBC2.

IV. Control Structure

The control block of the complete system is shown in Fig. 7. The grid voltage, the motor currents and the dc link voltage are the sensed quantities. These sensed quantities are transformed into dq reference frame using Clarke and Park transformation. The grid angle θ is calculated using PLL by aligning the grid voltage vector along the q-axis as shown in Fig. 7.

The system consists of an outer voltage control loop to maintain a constant dc link voltage. The voltage controller provides the reference current (corresponding to real power) for the two inner current loop for the respective IBCs based on the load sharing factor k_{lf}. If the currents are equally divided k_{lf} will be equal to 0.5.

The real power current component (I_{q_m} and I'_{q_m}) and reactive power current component (I_{d_m} and I'_{d_m}) are separately controlled using PI controllers for both IBCs. The zero sequence current is common for IBC1 and IBC2. Hence a single PI controller is used to control the zero sequence current.

V. Simulation Results

The proposed parallel combination of IBCs is simulated in MATLAB for a 1 kW system with a dc link voltage of 300 V and switching frequency of 10 kHz. The simulation results of the proposed R-Y-B/Y-R-B combination of IBCs is compared with conventional R-Y-B/R-Y-B combination as shown in Fig. 8. From the results it is clear that the currents of IBC1 and IBC2 are not balanced, but the grid current is balanced when the IBCs are paralleled using the proposed R-Y-B/Y-R-B configuration.

Fig. 8(a) shows the simulation result of R-Y-B/Y-R-B configuration. The phase currents of IBC1 and IBC2 are seen to be distorted, but the grid current which is sum of the currents through IBC1 and IBC2 are found to be balanced. Fig. 8(b) shows the simulation result of grid currents for R-Y-B (IBC1)/R-Y-B (IBC 2) configuration. The phase currents of IBC1 and IBC2 as well as the grid currents are found to be unbalanced.

The negative sequence component of phase currents of IBC1, IBC2 and grid current is taken for both configurations and compared in Table I. It can be seen that for R-Y-B/R-Y-B configuration, the negative sequence components gets added up in the grid currents, whereas for R-Y-B/Y-R-B configuration, the negative sequence terms are getting canceled in the grid currents.

TABLE I
Negative Sequence in R-Y-B/R-Y-B Configuration

conf	RYB-RYB				*Proposed RYB-YRB*			
System	Id		Iq		Id		Iq	
	mag	∠	mag	∠	mag	∠	mag	∠
IBC1	3.87	270	3.06	270	3.14	270	3.21	270
IBC2	3.87	270	3.06	270	2.14	90	2.37	90
grid	7.73	270	6.13	270	0.99	270	0.84	270

VI. Experimental Results

A. Experimental Setup

An experimental prototype was made with two 11 kW, 415 V three-phase induction machines which are rewound as split-phase machines. Two three-phase inverters made with SKM75GB12T4 IGBT half bridge modules are used as front end converters with a common dc link capacitor.

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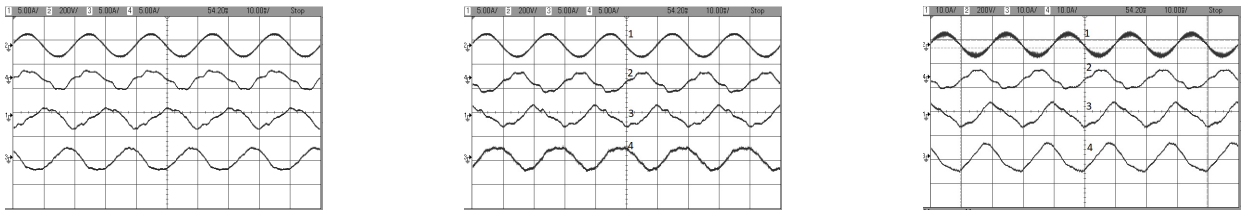
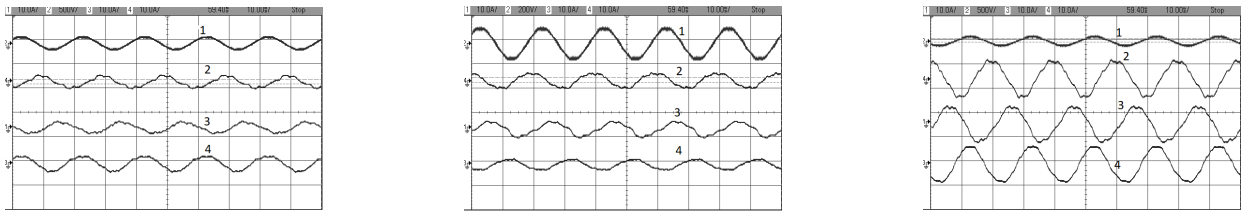


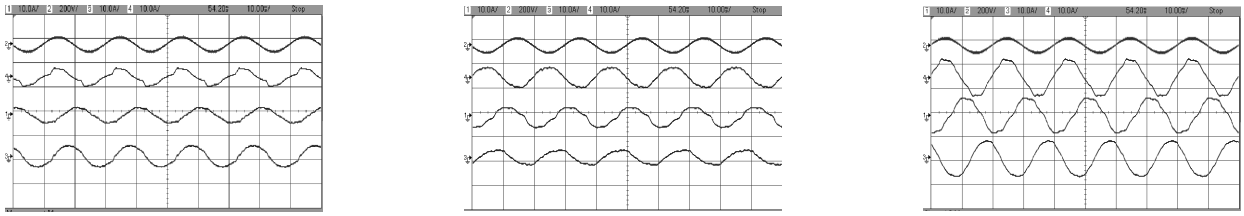
(a) (b) (c)

Fig. 9. Experimental results: steady state current waveforms: R-Y-B/R-Y-B configuration (unbalanced system). 1: Grid A phase voltage (200 V/div), 2: phase A current (5 A/div), 3: phase B current (5 A/div), and 4: phase C current (5 A/div). (a) IBC1 currents (R-Y-B configuration). (b) IBC2 currents (R-Y-B configuration). (c) Grid currents.



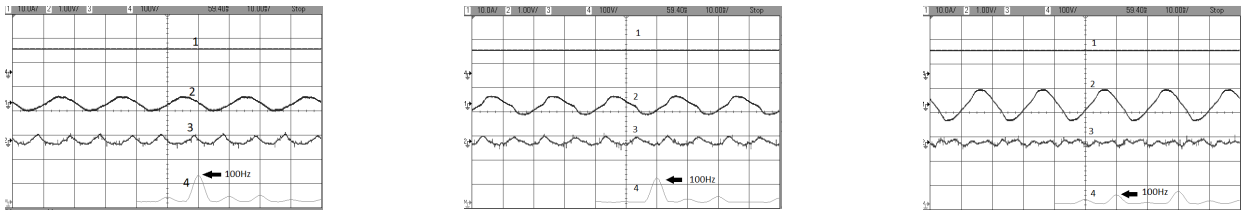
(a) (b) (c)

Fig. 10. Experimental results: steady state results at 300 V and 1 kW (proposed R-Y-B/Y-R-B configuration). 1: Grid C phase voltage (500 V/div), 2: A phase current (10 A/div), 3: B phase current (10 A/div), and 4: C phase current (10 A/div). (a) IBC1 currents (R-Y-B configuration). (b) IBC2 currents (Y-R-B configuration). (c) Grid currents.



(a) (b) (c)

Fig. 11. Experimental results at 500 W for the proposed R-Y-B/Y-R-B configuration. 1: Grid A phase voltage (200 V/div), 2: phase A current (10 A/div), 3: phase B current (10 A/div), and 4: phase C current (10 A/div). (a) IBC1 currents. (b) IBC2 currents. (c) Grid currents.



(a) (b) (c)

Fig. 12. Experimental results: reduction in unbalance. 1: DC link voltage (100 V/div), 2: Y phase current (10 A/div), 3: IQ current (10 A /div), and 4: harmonic spectrum centered around 50 Hz (100 mV/div). (a) IBC1-iq current and its harmonic spectrum. (b) IBC2-iq current and its harmonic spectrum. (c) Grid-iq current and its harmonic spectrum.

TMS320F28377S digital signal controller is used for controlling the proposed parallel system.

B. Steady State Results

1) R-Y-B/R-Y-B Configuration: Fig. 9 shows the steady state current waveforms for a power level of 500 W. It can be seen that the phase currents of IBC1, IBC2 and the grid currents are not balanced. The rms values of R phase current is 2.6 A, Y-phase current is 3.1 A and B-phase current is 3.9 A.

2) R-Y-B/Y-R-B Configuration: Fig. 10 shows the steady state current waveforms for a power level of 1 kW and 300 V.

TABLE II

RMS VALUE OF CURRENTS THROUGH IBC1, IBC2, AND GRID

Load	600 W and 232 V			1 kW and 300 V		
	$I_R(A)$	I_Y(A)	I_B(A)	$I_R(A)$	I_Y(A)	I_B(A)
IBC1	2.36	2.21	3.24	1.73	1.61	2.25
IBC2	2.96	2.97	2.17	2.21	2.19	1.55
GRID	5.31	5.20	5.41	5.28	5.12	5.28
THD(Grid)	8.26%	7.35%	5.37%	7.12%	7.28%	4.68%

The currents in IBC1 and IBC2 are unbalanced, but the grid current is found to be almost balanced and sinusoidal in shape.

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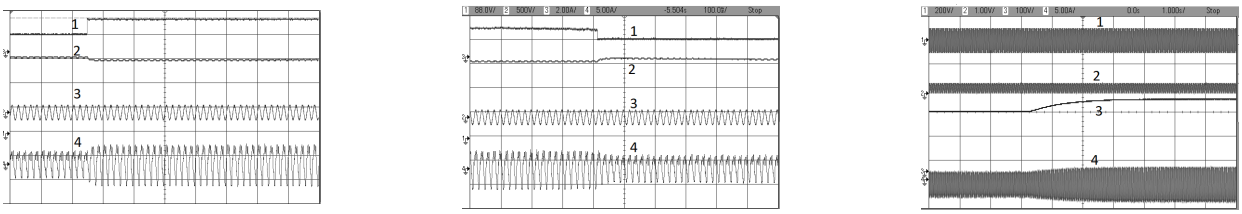

(a) (b) (c)

Fig. 13. Experimental results: transient results: for sudden change in load (a) from 400 to 800 W and (b) from 800 to 400 W [1: load current (2 A/div), 2: dc link voltage (88 V/div), 3: grid voltage (500 V/div), and 4: grid current (5 A/div)]. (c) Sudden change in dc link voltage from 250 to 300 V [1: grid angle θ, 2: grid voltage (200 V/div), 3: dc link voltage (100 V/div), and 4: grid current (5 A/div)].

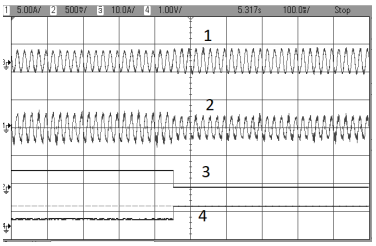

Fig. 14. Experimental results: sudden change from balanced power sharing to unbalanced power sharing (85% and 15%). 1: IBC1 phase current (10 A/div), 2: IBC2 phase current (5 A/div), 3: IQ ref for IBC2 from dsp (5 A/div), and 4: IQ ref for IBC1 from dsp (10 A/div).

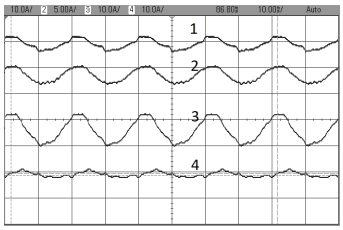

Fig. 15. Experimental results: circulating current at 320 V. 1: IBC1 phase A current (10 A/div), 2: IBC2 phase A current (10 A/div), 3: grid phase A current (10 A/div), and 4: 3*(zero sequence current) (5 A/div).

Fig. 11 shows the steady state current waveforms for 600 W power level for the same configuration. Table II gives the rms values of currents for 600 W and 1 kW power level. It can be seen that the rms values of IBC1 and IBC2 currents are not balanced, but the rms values of grid currents are almost equal. The current THD is for the grid currents are also given in the table. Fig. 12 shows the q-component of the current and its spectrum along with the dc link voltage. It shows that the q-component of current of IBC1 [see Fig. 12(a)] and IBC2 [see Fig. 12(b)] have a 100 Hz component along with the dc value due to the unbalance, but the 100 Hz component of grid current is canceled to a large extent, showing that the unbalance is minimized to a large extent using the proposed configuration.

C. Transient Results

The experimental results for sudden change in load current are shown in Fig. 13(a) and (b). The results for sudden change in dc link voltage is shown in Fig. 13(c) for the proposed configuration. Fig. 14 shows the results of current waveforms when there is a sudden change in load sharing between two IBCs from equal power sharing to unequal power sharing. During unequal power sharing IBC1 takes 85% load and IBC 2 takes 15% load. Using the zero sequence controller, the circulating current in the system is minimized as seen from the experimental result of Fig. 15.

TABLE III

ESTIMATED EFFICIENCY OF THE IBC

Component	Expt:	Sim:
Load	1 kW	10 kW
Inverter Loss (IBC 1)	3.657 W	42
Inverter Loss (IBC 2)	3.528 W	41
Rotor copper loss (Rr = 1.97Ω)(IBC 1)	13.22 W	168
Rotor copper loss (Rr = 1.97Ω)(IBC 2)	11.91 W	168
Stator copper loss (Rs= 1.05Ω) (IBC 1)	11.17 W	370
Stator copper loss (Rs= 1.05Ω) (IBC 1)	12.68 W	354
Efficiency	94.3%	88.5%

D. Calculation of Efficiency

The estimated efficiency of the IBC is calculated by taking into account, the inverter conduction and switching losses [14], stator and rotor copper loss of machine. The switches used for the inverters in the experimental set up is IGBT SKM75GB12T4, which has a rise time of 39 ns, fall time of 66 ns, and rated collector current of 75 A. The currents given in Table II for a load of 1 kW is used to calculate the losses, assuming modulation index to be 0.8. All the losses for both IBCs are tabulated in Table III. The calculated efficiency for a load of 1 kW is around 94%. Simulation using MATLAB was done for a load of 10 kW at dc link voltage of 800 V. The efficiency was found to be around 88.5%. The various losses are tabulated for 10 kW in Table III and the efficiency was found to be around 88.5%.

E. V2G Operation Results

Energy stored in the batteries of EVs can be used for intermittent power production, especially in applications like reactive power support, active power regulation, frequency regulation and current harmonic filtering [15]. The proposed parallel IBC is capable of handling power in both directions. Simulation [see Fig. 16(a)] and experimental results [see Fig. 16(c)] are included to verify the power flow from and to the grid. In Fig. 16(b), experimental results showing the change in mode of operation from grid to vehicle (G2V) and Vehicle to grid (V2G) is shown.

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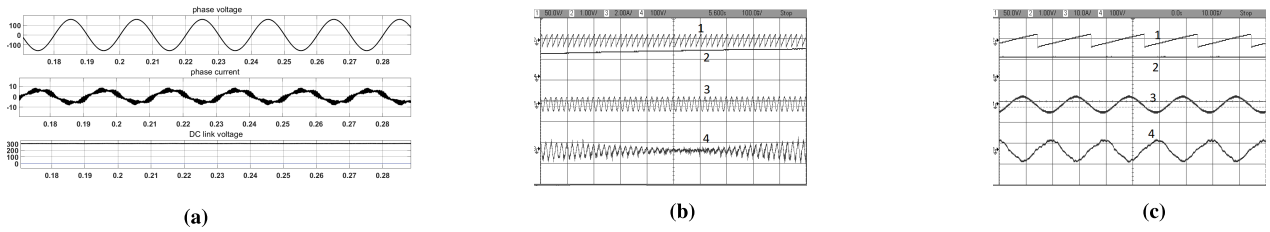

Fig. 16. (a) Simulation results. 1: Grid phase voltage, 2: grid phase current, and 3: dc link voltage. (b) and (c) Experimental results. 1: Grid angle θ from PLL, 2: dc link voltage (100 V/div), 3: grid phase voltage (50 V/div), and 4: grid phase current (2 A/div).

VII. Conclusion

A high power IBC for AWD EVs is proposed in this article. Two IBCs using split phase machine are connected in parallel to enhance the power handling capacity of the on-board charger. This is mainly used for charging the batteries in AWD EVs, where the existing driving hardware can be used for charging purpose. By connecting two IBCs in parallel, the power rating of the charger is improved and a very fast on-board charger is attained. A new configuration, namely R-Y-B/Y-R-B configuration is developed which reduced the unbalance created in a single IBC using split phase machine. The zero sequence model of the system is derived and the equation for zero sequence circulating current is found. A zero sequence controller is used to limit the circulating current. Thus, a high power, unity power factor on board charger with zero instantaneous torque and balanced currents is developed in this article. Experimental results are included which verified the theoretical analysis done in this work. The results prove that the proposed configuration of parallel IBCs and the proposed control technique is a viable on-board charging scheme for AWD vehicles.

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